

Thermochemistry

Experiment No. 7

Date: / /

Aim: To determine enthalpy of dissolution of copper sulphate ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$) in water at room temperature.

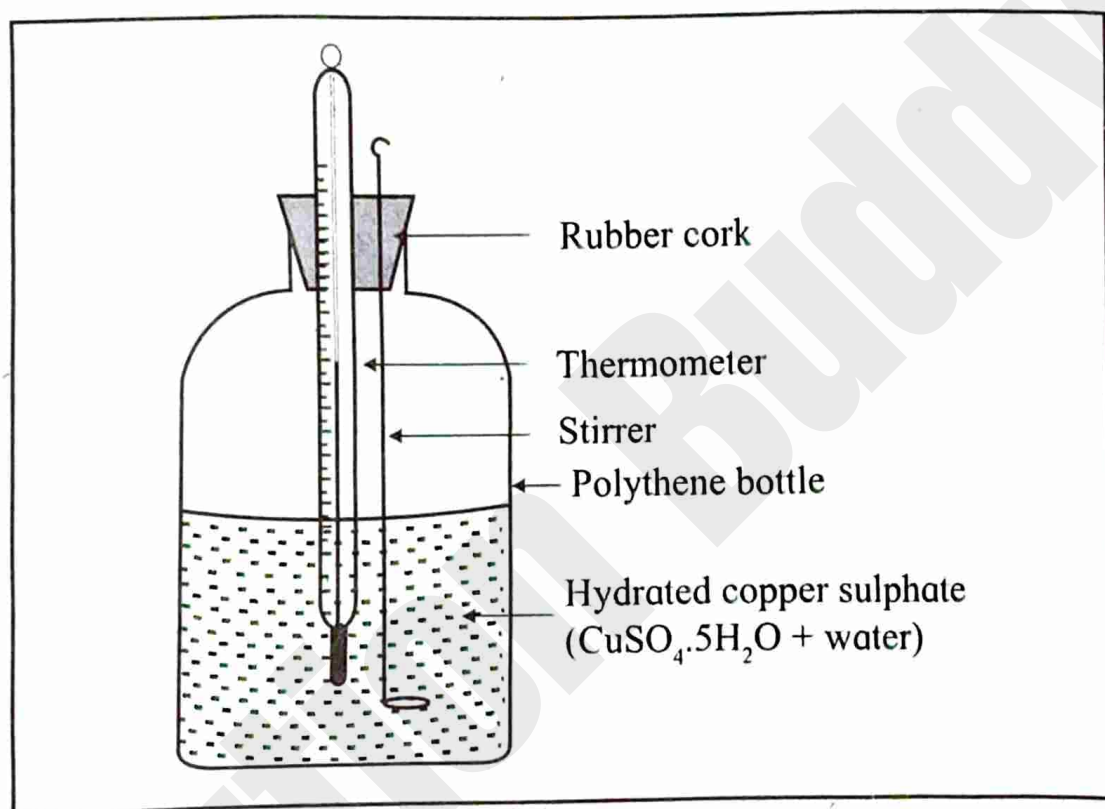
Theory: Heat of solution is the change in enthalpy, when one mole of a substance is dissolved in a large amount of solvent so that further dilution does not change the enthalpy. Generally for 1 mole of solute 400 moles of solvent is recommended. Heat of solution is due to ionization or some hydrate formation. Dissolution may be exothermic or endothermic process.

Reaction: $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}(\text{s}) + \text{H}_2\text{O}(\text{l}) \longrightarrow \text{CuSO}_4(\text{aq})$; $\Delta H = + 11.7 \text{ kJ/mol}$

Apparatus: Polythene bottle, cork having two holes, two 100 mL beakers, measuring cylinder.

Chemicals: Copper sulphate and distilled water.

Diagram:



Procedure:

1. Take 25mL water in a polythene bottle.
2. Fit cork on polythene bottle, insert stirrer and thermometer in two holes of the cork.
3. Record the initial temperature of water ($t_1^\circ\text{C}$).
4. Add 0.5 g of powdered copper sulphate ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$) to the polythene bottle.
5. Stir the solution to dissolve the copper sulphate completely.
6. Record the final temperature, when substance just dissolves ($t_2^\circ\text{C}$).

Note :

- i. Specific heat capacity of commonly used PET type of polythene bottle can be approximated to $1 \text{ J/g } ^\circ\text{C} = 1000 \text{ J/kg } ^\circ\text{C} \approx 0.25 \text{ cal/g } ^\circ\text{C}$
- ii. If the bottle is thin walled, its mass can be neglected if mass of water inside it is more than 100g.
- iii. If a heavy bottle of different material is used, we need to know and use its water equivalent.

Observation Table:

Sr. No.	Observation	Symbol	Value
1	Mass of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ taken	–	0.5 g
2	Volume of water	–	25 mL
3	Mass of solution	m	25.5 g
4	Initial temperature water	t_1	26.0 °C
5	Final temperature of the solution	t_2	26.5 °C
6	Change in temperature	$\Delta t = t_2 - t_1$	$\Delta t = 26.5 - 26.0$ $= 0.5 \text{ °C}$
7	Specific heat capacity of water	S	4.184 J/g °C

Calculation: Heat absorbed for dissolution of 0.5 g of CuSO_4

$$Q = m S \Delta t$$

$$Q = 25.5 \times 4.184 \text{ J/g °C} \times 0.5 \text{ °C}$$

$$Q = 53.346 \text{ J}$$

Molar mass of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is 249 g/mol

$$\begin{aligned} \text{Moles of } \text{CuSO}_4 \text{ dissolved} &= \frac{\text{Mass of } \text{CuSO}_4}{\text{Molar mass of } \text{CuSO}_4} \\ &= \frac{0.5}{249} \\ &= 2 \times 10^{-3} \text{ mol} \end{aligned}$$

..... (Q) is the heat absorbed, when 2×10^{-3} mol of CuSO_4 is dissolved

The enthalpy of solution of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is therefore

$$\Delta H = \frac{Q}{2 \times 10^{-3}}$$

$$\Delta H = \frac{53.346}{2 \times 10^{-3}} \text{ J/mol}$$

$$\Delta H = 26673 \text{ J/mol}$$

$$\Delta H = +26.67 \text{ kJ/mol}$$

Since dilute solution is used
density of solution is 1 g/mL
Hence 25.5 mL = 25.5 g

Space for log calculation

Number	log

Result:

The enthalpy of dissolution of given copper sulphate is**+26.67**..... kJ/mol.

Remark and sign of teacher:

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MCQ

Select the correct answer for the following multiple choice type questions.

1. The removal of one electron from each atom in one mole is called....

a. enthalpy
✓. first ionization enthalpy

c. second ionization enthalpy
d. electronegativity
2. Mathematically the enthalpy is represented by equation

a. $\Delta H = H_2 - H_1$
b. $H = U + PV$

c. $\Delta H = \Delta U$
✓d. $\Delta H = \Delta U + P \Delta V$
3. The amount of copper sulphate used in dissolution is

a. 0.5 g
b. 2 g
c. 3 g
✓. 7 g
4. The dissolution of copper sulphate is.....

a. exothermic reaction
✓. endothermic reaction

c. reversible reaction
d. irreversible reaction

Short answer questions

1. What is enthalpy?

Ans. The enthalpy of a system (defined as sum of internal energy of the system and the energy that arises due to its presence and volume). $H = U + PV$

2. What is enthalpy of the solution?

Ans. The enthalpy of solution is defined as enthalpy change in a process in which one mole of a substance is dissolved in a specified quantity of a solvent so as to form solution of concentration.

3. Write two reactions each showing exothermic and endothermic nature.

Ans. Exothermic reactions: (1) $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$ $\Delta H = - 572 \text{ kJ}$

Endothermic reactions: (1) $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$ $\Delta H = + 178 \text{ kJ}$

4. Why is enthalpy of solution for some substances negative while for others it is positive?

Ans. When salt is dissolved in water or solvent to form solution, if heat is absorbed the reaction will be endothermic (positive ΔH) and if heat is evolved the reaction will be exothermic (negative ΔH).

5. Why is copper sulphate taken in powdered form?

Ans. Copper sulphate is a pentahydrate crystals in powdered form and exist in nature which makes it easy to take part in the reaction.

Remark and sign of teacher:

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